

EXACT ABSORPTION AND CLOCK LAWS FOR RANDOM MAJORITY TRIPLE CONTRACTION

ANONYMOUS

ABSTRACT. Choose a consecutive triple of a binary word uniformly, replace it by its majority bit, and repeat until one bit remains. On a two-run word $0^a 1^b$ of odd length, the two-run family is closed and the terminal-one probability is $(b-1)/(a+b-2)$. We enumerate the terminal histories, derive a marked recursion for the number of cross-boundary contractions, and obtain a reciprocal formula for the event that exactly one such contraction occurs. With independent rate-one clocks on current windows, the entire vector of holding times is independent of the embedded window history. The completion time is an odd-rate hypoexponential sum; its exponential is exactly beta distributed and has a gamma scaling limit. The discrete contraction count is deterministic and is kept separate from this continuous elapsed time.

1. POSITION AND MODEL

Majority-rule opinion dynamics on fixed populations is classical; for example, Krapivsky and Redner [2003] update groups of agents without shrinking the population. Majority automata likewise retain a fixed carrier [Goles et al., 2016]. We give both bodies of work full ownership of the majority operation and its consensus interpretation. Our deliberately narrow object is instead the following length-reducing word process.

Let $w = w_1 \cdots w_n \in \{0, 1\}^n$, where n is odd. At length $\ell > 1$, choose one of the $\ell - 2$ consecutive length-three windows uniformly and replace its three bits by their majority. Thus the length falls by two, and absorption at one bit occurs after the deterministic number $(n - 1)/2$ of contractions. A *history* records the successive window positions in the current, shortened words. It is not an unlabelled contraction tree.

We focus on $0^a 1^b$ with $a, b \geq 1$ and odd $n = a + b$. A contraction is *crossing* if its window is heterogeneous. Because the word has two runs at every step below, this is equivalent to crossing the current run boundary. We use $(-1)!! = 1$.

2. TWO-RUN CLOSURE AND ABSORPTION

Proposition 2.1 (Closed two-run kernel). *From state (a, b) the only successors and their window multiplicities are*

$$(1) \quad \begin{array}{c|c} \text{successor} & \text{multiplicity} \\ \hline (a-2, b) & \max\{a-2, 0\}, \\ (a, b-2) & \max\{b-2, 0\}, \\ (a-1, b-1) & \mathbf{1}_{\{a \geq 2\}} + \mathbf{1}_{\{b \geq 2\}}. \end{array}$$

The displayed multiplicities sum to $a + b - 2$. In particular, the two-run family, allowing a zero-size terminal block, is closed.

Proof. A 000 window removes two zeros and a 111 window removes two ones. There are $a - 2$ and $b - 2$ such windows when the corresponding block has size at least three. At the boundary, 001 exists exactly when $a \geq 2$ and contracts to 0, while 011 exists exactly when $b \geq 2$ and contracts to 1; either operation removes one bit from each block. These cases partition all current windows, and direct addition gives $a + b - 2$. $\square$

Theorem 2.2 (Endpoint and history laws). *For $a, b \geq 1$ and odd $n = a + b$,*

$$(2) \quad \Pr_{a,b}(1) = \frac{b-1}{n-2}, \quad \Pr_{a,b}(0) = \frac{a-1}{n-2}.$$

Moreover, among the $(n-2)!!$ complete window-position histories,

$$(3) \quad \#\{h : h \rightarrow 1\} = (b-1)(n-4)!!, \quad \#\{h : h \rightarrow 0\} = (a-1)(n-4)!!.$$

Proof. Put $p(0, b) = 1$ and $p(a, 0) = 0$. The boundary cases $a = 1$ and $b = 1$ are forced: homogeneous contractions reduce the non-singleton block until its boundary window absorbs the singleton of the opposite type. Hence $p(1, b) = 1$ and $p(a, 1) = 0$, in agreement with (2).

For $a, b \geq 2$, use the transition multiplicities in Proposition 2.1. Substitution of $p(a, b) = (b-1)/(a+b-2)$ in the first-step recurrence, using the forced boundary values when a successor block has size zero or one, gives

$$p(a, b) = \frac{1}{n-2} \sum_{(a', b')} m((a, b), (a', b')) p(a', b').$$

After multiplying by $(n-2)(n-4)$, both sides reduce to $(b-1)(n-4)$; terms whose multiplicity is zero are absent. Induction on odd n proves the first formula, and complementation proves the second.

At successive lengths the numbers of available windows are $n-2, n-4, \dots, 1$. Thus every complete position history has the common probability $1/(n-2)!!$, and there are $(n-2)!!$ histories. Multiplying (2) by this denominator proves (3). $\square$

3. THE MARKED CROSS-BOUNDARY LAW

For a history h , let $C(h)$ be its number of crossing contractions. Define the history-count polynomials

$$H_{a,b}^\varepsilon(u) = \sum_{h: h \rightarrow \varepsilon} u^{C(h)}, \quad \varepsilon \in \{0, 1\}.$$

Theorem 3.1 (Cross-count PGF before normalization). *For $a, b \geq 1$ with $a+b$ odd,*

$$(4) \quad \begin{aligned} H_{a,b}^1(u) &= \mathbf{1}_{\{a \geq 3\}}(a-2)H_{a-2,b}^1(u) + \mathbf{1}_{\{b \geq 3\}}(b-2)H_{a,b-2}^1(u) \\ &\quad + (\mathbf{1}_{\{a \geq 2\}} + \mathbf{1}_{\{b \geq 2\}})uH_{a-1,b-1}^1(u), \end{aligned}$$

with $H_{0,b}^1(u) = (b-2)!!$ on the reachable odd one-run boundary, $H_{a,0}^1(u) = 0$, and

$$(5) \quad H_{a,b}^0(u) = H_{b,a}^1(u).$$

If $b \geq 2$, the coefficient support of $H_{a,b}^1$ is exactly

$$(6) \quad \{c : 1 \leq c \leq \min(a, b-1), c \equiv a \pmod{2}\};$$

if $b = 1$, this polynomial is zero. The probability PGF conditional on terminal one is $H_{a,b}^1(u)/H_{a,b}^1(1)$ whenever $b \geq 2$.

Proof. Marking the third transition of (1) by u gives (4); the one-run boundary and bit-complement symmetry give the remaining identities. On a terminal-one history, each homogeneous-zero contraction removes two zeros and each crossing contraction removes one. Hence $C \equiv a \pmod{2}$ and $C \leq a$. Each crossing also removes one one, so at least one one remaining at absorption gives $C \leq b-1$. A crossing is necessary when $a > 0$.

It remains to see that every allowed coefficient is positive. Induct on $a+b$. If an allowed c satisfies $c \leq a-2$, first use a homogeneous-zero window and apply the induction hypothesis to $(a-2, b)$. Otherwise parity forces $c = a$. If $a = 1$ (and hence b is even), a crossing $011 \mapsto 1$ leads directly to the odd one-run boundary $(0, b-1)$ and contributes the positive coefficient of u . If $a \geq 2$, a crossing transition leads to $(a-1, b-1)$, where $c-1 = a-1 \leq b-2$ satisfies the inductive support conditions. The transition multiplicities are positive, completing the induction. $\square$

Corollary 3.2 (Exactly one crossing). *For odd $n = a+b$,*

$$(7) \quad \Pr_{a,b}(\text{terminal } 1, C = 1) = \begin{cases} 1/a, & a \text{ odd}, \\ 0, & a \text{ even}, \end{cases}$$

$$(8) \quad \Pr_{a,b}(\text{terminal } 0, C = 1) = \begin{cases} 1/b, & b \text{ odd}, \\ 0, & b \text{ even}. \end{cases}$$

Equivalently, when a is odd, $[u]H_{a,b}^1(u) = (n-2)!!/a$, with the symmetric zero formula. Consequently the unconditional probability of exactly one crossing is the reciprocal of the unique odd block size.

Proof. The parity statement follows from (6). Suppose a is odd, so b is even, and write $q(a, b)$ for the probability on the left of (7). The case $a = 1$ has exactly one eventual crossing, hence $q(1, b) = 1$. For odd $a \geq 3$, a crossing at the first step leaves a positive even zero block, which cannot disappear using

only homogeneous triples; that branch cannot have total cross count one. The two homogeneous first steps, when present, and induction therefore give

$$q(a, b) = \frac{a-2}{n-2} \frac{1}{a-2} + \frac{b-2}{n-2} \frac{1}{a} = \frac{1}{a},$$

where a zero-multiplicity term is omitted. Complementing bits proves (8). Multiplication by the common history denominator proves the coefficient formula. Exactly one of a, b is odd, so the two terminal events are disjoint and only the corresponding term is nonzero. $\square$

4. CONTINUOUS TIME AND WHOLE-HISTORY INDEPENDENCE

Give every current length-three window an independent rate-one exponential clock and apply the first ring. After an update, use the standard Markov construction: new clocks are fresh, while surviving clocks have fresh residual exponentials by memorylessness. Let J_r be the winning current window at contraction r , E_r its holding time, and $\tau_n = \sum_r E_r$.

Theorem 4.1 (Clock/history separation). *For every odd initial word, the vector $(E_r)_{r=1}^{(n-1)/2}$ is independent of the complete embedded history $(J_r)_{r=1}^{(n-1)/2}$. Its coordinates are independent and*

$$(9) \quad \tau_n \stackrel{d}{=} \sum_{k=1,3,\dots,n-2} Y_k, \quad Y_k \sim \text{Exp}(k).$$

In particular, τ_n is independent of the terminal bit and of C .

Proof. At length ℓ , the minimum of the $\ell-2$ rate-one clocks is $\text{Exp}(\ell-2)$, its winning index is uniform, and the two variables are independent. Conditional on the complete past, strong Markov memorylessness restarts the same statement at deterministic length $\ell-2$. Induction over the deterministic length sequence proves the product factorization of the whole holding-time vector and the whole winner vector. Summing the holding times gives (9); the endpoint and C are functions of the winner history. $\square$

Write $n = 2m + 1$. The next consequences concern elapsed time only; they are not statements about the fixed discrete count m .

Corollary 4.2 (Exact transform, Beta law, and moments). *For $s \geq 0$,*

$$(10) \quad \mathbb{E}e^{-s\tau_n} = \prod_{j=0}^{m-1} \frac{2j+1}{2j+1+s}$$

$$(11) \quad = \frac{\Gamma(m + \frac{1}{2})\Gamma(\frac{1+s}{2})}{\Gamma(\frac{1}{2})\Gamma(m + \frac{1+s}{2})}.$$

Moreover,

$$(12) \quad e^{-2\tau_n} \stackrel{d}{=} B_m, \quad B_m \sim \text{Beta}(\frac{1}{2}, m),$$

and

$$(13) \quad \mathbb{E}\tau_n = \sum_{j=0}^{m-1} \frac{1}{2j+1}, \quad \text{Var}(\tau_n) = \sum_{j=0}^{m-1} \frac{1}{(2j+1)^2}.$$

Proof. Independence in (9) gives (10); separating factors of two and using $\Gamma(x+m)/\Gamma(x) = \prod_{j=0}^{m-1} (x+j)$ gives (11). For $q \geq 0$, substitute $s = 2q$ in that formula. The result is

$$\mathbb{E}(e^{-2\tau_n})^q = \frac{\Gamma(q + \frac{1}{2})\Gamma(m + \frac{1}{2})}{\Gamma(\frac{1}{2})\Gamma(m + q + \frac{1}{2})},$$

the moment formula of $\text{Beta}(1/2, m)$. Since both variables lie in $[0, 1]$, their moments determine their laws. Finally, means and variances add in (9). $\square$

Theorem 4.3 (Gamma scaling limit). *With the shape-rate convention,*

$$(14) \quad me^{-2\tau_n} \xrightarrow{d} G, \quad G \sim \text{Gamma}(\frac{1}{2}, 1).$$

Equivalently, $\tau_n - \frac{1}{2} \log m \xrightarrow{d} -\frac{1}{2} \log G$.

Proof. Use the beta-gamma representation $B_m = X/(X + Y_m)$ with independent $X \sim \text{Gamma}(1/2, 1)$ and $Y_m \sim \text{Gamma}(m, 1)$. The law of large numbers gives $Y_m/m \rightarrow 1$ in probability, while $X/m \rightarrow 0$. Hence $mB_m = X/[X/m + Y_m/m] \rightarrow X$ in distribution. Apply Corollary 4.2; the logarithmic statement follows by the continuous mapping theorem. $\square$

5. EXACT CONTROLS AND LIMITS OF SCOPE

The paper-local verifier uses only integer and rational arithmetic. It does not sample. Table 1 records its finite falsification grid.

TABLE 1. Exact-arithmetic control envelope.

control lane	checked cells or inputs
all binary words, odd length at most 15	43,690
two-run inputs, odd length at most 201	10,100
marked cross-polynomial coefficients, length at most 101	90,950
clock product-transform cells	1,000
exact assertions	190,740

Literal marked word histories are also compared through length 13, and the joint transform factorization from Theorem 4.1 is checked on every word through length 11. These bounded controls are not proofs of the all-length statements or of the limit, and they provide no novelty evidence.

We have not derived a comparable closed atlas for arbitrary multi-run words. The result is therefore a special invariant-family calculation, not a general solution of random majority contraction. The majority operation, standard clock construction, beta–gamma identities, and convergence tools receive no credit. The exact conjunction above remains an internal theorem package; external circulation is on hold pending a deeper owner search.

REFERENCES

- Eric Goles, Pedro Montealegre, Ville Salo, and Ilkka Törmä. PSPACE-completeness of majority automata networks. *Theoretical Computer Science*, 609(1):118–128, 2016. doi: 10.1016/j.tcs.2015.09.014.
- P. L. Krapivsky and S. Redner. Dynamics of majority rule in two-state interacting spin systems. *Physical Review Letters*, 90(23):238701, 2003. doi: 10.1103/PhysRevLett.90.238701.